The sharp A_p bound for the Hardy–Littlewood maximal operator Simon Gustafsson

Notation. *Cubes are closed with sides parallel to the axes, $\ell(Q)$ is the side length of Q, and $3Q$ has the same centre as Q and side length $3\ell(Q)$. Centred cubes are $Q(x,r) := \{y : |y-x|_\infty \le r/2\}$, and $v(E) := \int_E v\,dy$.*

Definition 1 (Weights)**.** *A* weight *is a function $w \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$ with $w \ge 0$ that is not almost everywhere zero.*

Definition 2 (Weighted L^p spaces)**.** *Let w be a weight and let $1 \le p < \infty$. The weighted L^p space is*

$$L^p_w(\mathbb{R}^d) := \left\{ f : \mathbb{R}^d \to \mathbb{R} \text{ measurable} : \int_{\mathbb{R}^d} |f(y)|^p\, w(y)\, dy < \infty \right\},$$

with norm $\|f\|_{L^p_w} := \left(\int_{\mathbb{R}^d} |f(y)|^p\, w(y)\, dy\right)^{1/p}$. Moreover, $L^\infty_w(\mathbb{R}^d)$ is the space of measurable functions f with $\|f\|_{L^\infty_w} := \inf\{t \ge 0 : w(\{|f| > t\}) = 0\} < \infty$.

Definition 3 (Maximal operators)**.**

$$M_w f(x) := \sup_{Q \ni x} \frac{1}{w(Q)} \int_Q |f|\, w\, dy, \qquad M^c_w f(x) := \sup_{r>0} \frac{1}{w(Q(x,r))} \int_{Q(x,r)} |f|\, w\, dy.$$

For $w \equiv 1$ these are M and M^c.

Definition 4 (A_p characteristic)**.**

$$[w]_{A_p} := \sup_Q \left(\frac{1}{|Q|}\int_Q w\, dy\right)\left(\frac{1}{|Q|}\int_Q w^{-\frac{1}{p-1}}\, dy\right)^{p-1},$$

and $w \in A_p$ if $[w]_{A_p} < \infty$.

Definition 5 (Dual weight)**.** *$\sigma := w^{-\frac{1}{p-1}}$, so that $\sigma^{1-p} = w$, $w^{1-\frac{p}{p-1}} = \sigma$, $\sigma^p w = \sigma$ and*

$$[w]_{A_p} = \sup_Q \frac{w(Q)}{|Q|}\left(\frac{\sigma(Q)}{|Q|}\right)^{p-1}.$$

Theorem 2 (Buckley 1993, Lerner 2008)**.** *Let $1 < p < \infty$.*

(i) There is a constant $D_{p,d}$, depending only on p and d, such that for every $w \in A_p$ and every $f \in L^p_w(\mathbb{R}^d)$,

$$\|Mf\|_{L^p_w} \le D_{p,d}\, [w]_{A_p}^{\frac{1}{p-1}}\, \|f\|_{L^p_w}.$$

(ii) If $\alpha < \frac{1}{p-1}$, then there is no constant D such that $\|Mf\|_{L^p_w} \le D\, [w]^\alpha_{A_p}\, \|f\|_{L^p_w}$ for all $w \in A_p$ and all $f \in L^p_w(\mathbb{R}^d)$.